Student ID _____ Student Name _____

University of the South Pacific

Faculty of Science, Technology and Environment

School of Computing, Information and Mathematical Sciences

IS221 Web Applications Development

Mid Semester Test

Semester 1, 2022

Possible Marks:	**100**
Weight:	**20%**
Time Allowed:	**110** minutes
Number of Pages:	**11** (including this page)

Learning Outcomes:

2. Apply ethical practices to the development of the websites

Instructions:

1. Sections A, B, C, D: Answer ALL questions.
2. Write ALL the answers in the spaces provided in the Answer Booklet.
3. If you require extra space for your answers, use the last page.

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Section A: Multiple Choice 10 Marks

Instructions:
Identify and circle the choice that best completes the statement or answers the question. There is only **one** possible answer.

1. ____ is a subset of SGML, but has no predefined elements such as <h1> or <p>.

   a. XML.
   b. HTML.
   c. XHTML.
   d. AML.

2. The benefit of the XHTML ____ type is that it allows you to gradually migrate from existing HTML code that may still contain font and display information to the more syntactically correct, cleaner markup code necessary to match the strict type.

   a. Strict
   b. Frameset
   c. Transitional
   d. Partition

3. The ____ for interpreting the HTML tags varies from browser to browser, resulting in potentially conflicting interpretations of the way the HTML file is displayed.

   a. programming
   b. design
   c. coding
   d. logic

4. If the ____ you choose is not installed on the user's machine, the Web page will appear in a default typeface.

   a. graphic text
   b. text
   c. font
   d. type layout

5. Given the URL: http://www.usp.ac.fj/IS221/lab4.html, IS221is the _____part of the URL.

   a. Domain Name
   b. Path Name
   c. Protocol
   d. File Name

6. To create a style that will be applied to the contents of a paragraph which is to be colored green, with font-size of 20 pixels and right aligned, the following style should be used:

   a. p{ color: green; font-size: 20px; align:right; }
   b. p{ color= " green"; font-size = "20px"; align="right"; }
   c. p{ color: green; font-size: 20%; align:right; }
   d. p{ color: green; font: 20px; align:right; }

7. To create a table with a border width of 1, margin width of 1 between cells and cells that take up 3 columns, the following tags should be used:

   a. <table border = "1" >, <table cellpadding = "1" >, <table rowspan = "3" >
   b. <table border = "1" >, <table cellspacing = "1" >, <table colspan = "3" >
   c. <table border = "1" >, <table cellpadding = "1" >, <rowspan = "3" >
   d. <table border = "1" >, <table cellspacing = "1" >, <colspan = "3" >

8. Moodle is a fundamental aspect of USP's e-learning system and employs the use of _____ and _____ aspart of its content goal.

   a. Special interest, portal
   b. Publishing, portal
   c. Portal, Intranet/Extranet
   d. Publishing, intranet/extranet

9. The site structure used by USP's website is _____ due to the size of the site and extensive content collection.

   a. Web
   b. Tutorial
   c. Catalog
   d. Hierarchical

10. John would like to register a domain name for his site that will be a hub of information for transgender individuals all over Fiji. An appropriate domain name could be:

    a. www.transgender.org.fj
    b. www.transgender.com.fj
    c. www.transgender.ac.fj
    d. www.transgender.net.fj

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Section B: Matching Questions 10 Marks

Write the letter of the correct term that matches the given description in the space provided

A. HTML5

*B. *

C. Perceivability

D. Compatibility

E. Maintainability

F. Portability

G. Portal

H. Operability

I. <! Doctype html>

J. FTP

1. ___E_____ Advantage of using CSS.

2. ___A_____ Less attributes due to CSS.

3. ___I_____ Forces the browser to display an HTML5 web page in standards mode.

4. ___B_____ Enables parts of text content to be treated differently from the surrounding text

5. ____G____ Logging in as a student on Moodle will give you a different view of it compared to a lecturer's view of Moodle..

6. ___H_____ 7 year old is able to login to Facebook and view their cousin's birthday photos

7. ___C_____ Users visiting Vodafone Fiji's website instantly know what the website is about

8. ____J____ Uploading of completed website to hosting server.

9. ___F_____ Being able to view Digicel Fiji's website on an iPhone, Samsung Galaxy mobile phone and iPad Air

10. ____D____ Facebook page rendered with minimal differences across Chrome, Firefox and Internet Explorer.

Section C: Coding Questions 30 Marks

1) The code below in Figure 1 represents the content for the home page of a website. using appropriate HTML5 semantic elements markup the code in Figure 1 below to make meaningful semantic HTML5 code.

[10 Marks]

```
<!DOCTYPE HTML>
<html>
<head>
<meta charset="utf-8">
<title>Welcome to IS221</title>
</head>
<body>
<div id="pageHeader">Header content goes here.</div>
<div id="mainNav">Navigation goes here.</div>
<div id="content">
<div class="banner">Banner content goes here</div>
<div id="home">Article content goes here</div>
<div id="sidebar">Sidebar content goes here</div>
</div>
<div id="pageFooter">Footer content goes here</div>
</body>
</html>
```

Figure 1: Home Page

```
<!DOCTYPE HTML>
<html>
<head>
<meta charset="utf-8">
<title>Welcome to IS221</title>
</head>
<body>
<header id="pageHeader">Header content goes here.</header>
<nav id="mainNav">Navigation goes here.</nav>
<section id="content">
<div class="banner">Banner content goes here</div>
<article id="home">Article content goes here</article>
<aside>Sidebar content goes here</aside>
</section>
<footer id="pageFooter">Footer content goes here</footer>
</body>
</html>
```

2) This question is all about **ASP.NET**. Most of the ASP.NET tags start with an asp: prefix. The given code is for a button in ASP.NET.

```
<asp:Button ID="Button1" runat="server" Text="Button" />
```

The code for the above button will be rendered in the browser as XHTML codes. Complete the code below to show the equivalent XHTML codes, by filling the appropriate values for each attribute:

```
<input type="___1___ " name="___2___ " value="___3__ " id="___4___ ">
```

[2 Marks]

ANSWER:

```
<input type="submit" name="Button1" value="Button" id="Button1">
```

3) The example below is that of a Page Directive which often appears on top of each page or Web Form. What is the main purpose of the Page Directive?

```
<%@ Page Language="VB" CodeFile="Login.aspx.vb" Inherits="Login" title="
Login Page" AutoEventWireup="false" %>
```

ANSWER: **[1 Mark]**

Set the default page level properties or a series of instructions that tell ASP.NET runtime how the page should behave.

4) What are the names of the Web Form and the corresponding code behind the page file In the example in question 3 above? **[2 Marks]**

Login.aspx
Login.aspx.vb

5) Figure 2 below shows a Web form that has two list controls. There is a button, when clicked, displays the selected items as text in a Label control. Part of the codes for the Web Form name as **ListControl.aspx** is also given below:

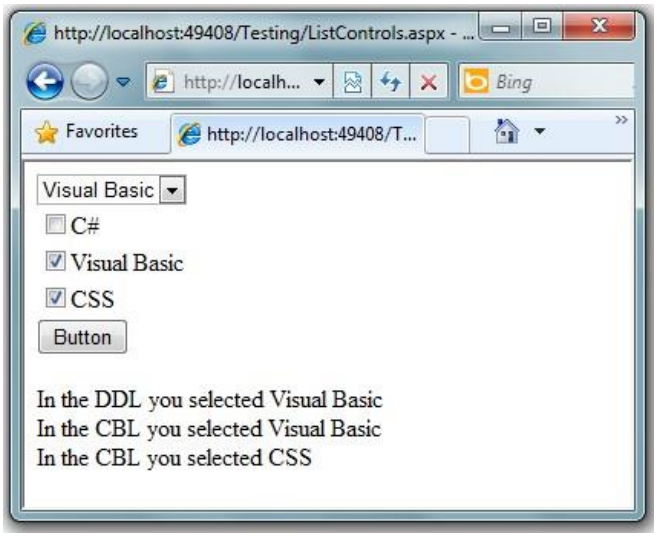


Figure 2:Web Form with two list controls

Part of the codes for the Web Form, **ListControl.aspx**, is given below. Some of the codes were removed for the purpose of this question.

```
<!--        Page Directive comes here                    -->
<%@ Page Language="VB" AutoEventWireup="false" CodeFile="ListControls.aspx.vb"
Inherits="ListControls" %>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <form id="form1" runat="server">
    <div>
         <!--        DropDownList Control                        -->
         <asp:DropDownList ID="DropDownList1" runat="server">
            <asp:ListItem>C#</asp:ListItem>
            <asp:ListItem>Visual Basic</asp:ListItem>
            <asp:ListItem>CSS</asp:ListItem>
        </asp:DropDownList>
         <!--        CheckBoxList Control comes here             -->

        <!--        Button Control                              -->

        <asp:Button ID="Button1" runat="server" Text="Button" />
 <br />
        <br />
         <!--        Label Control comes here                    -->

    </div>
    </form>
</body>
</html>
```

a) Write the ASP codes that should be placed under the **CheckBoxList control** comment in the `ListControl.aspx` Web Form source code that renders the above **three check boxes** in your browser. **[5 Marks]**

Answer :

```
<asp:CheckBoxList ID="CheckBoxList1" runat="server">
    <asp:ListItem>C#</asp:ListItem>
    <asp:ListItem>Visual Basic</asp:ListItem>
    <asp:ListItem>CSS</asp:ListItem>
</asp:CheckBoxList>
```

b) Write the ASP codes that should be placed under the **Label control** comment in the `ListControl.aspx` Web Form source code that renders the **comments** below the button. **[3 Marks]**

Answer :

```
<asp:Label ID="Label1" runat="server" Text="Label"></asp:Label>
```

c) Write the corresponding XHTML code for Label control that renders onto the browser and result in the three lines of labels under the button. **[3 Marks]**

Answer :

```
<span id="Label1">In the DDL you selected Visual Basic<br />
In the CBL you selected Visual Basic<br />
In the CBL you selected CSS<br /></span>
```

d) The codes that create the labels in part [c] were generated by Visual Basic .NET codes that were placed in the Click Event Handler for the button. Part of the codes of the Event Handler is given below. Identify the missing numbered properties or Visual Basic .NET codes.

```
Protected Sub ___1___(ByVal sender As Object, ByVal e As System.EventArgs) _
              Handles Button1.Click

<!--     The first label or line              -->

    Label1.___2___ = "In the DDL you selected " & DropDownList1.___3___  & _
                "<br />"

<!--     The second and third labels or lines              -->

    ___4___ item As ListItem In CheckBoxList1.___5___
        If item.___6___  Then
           Label1.___7___  &= "In the CBL you selected " & item.___8___ & _
                              "<br />"
        End If
    Next

End Sub
```

[4 Marks]

Answer :

1. Button1_Click
2. Text
3. SelectedValue
4. For Each
5. Items
6. Selected
7. Text
8. Value

Section D: Short Answer Questions 50 Marks

1) Given the source code below, with the help of a diagram illustrate how the web page is going to be rendered. **[10 Marks]**

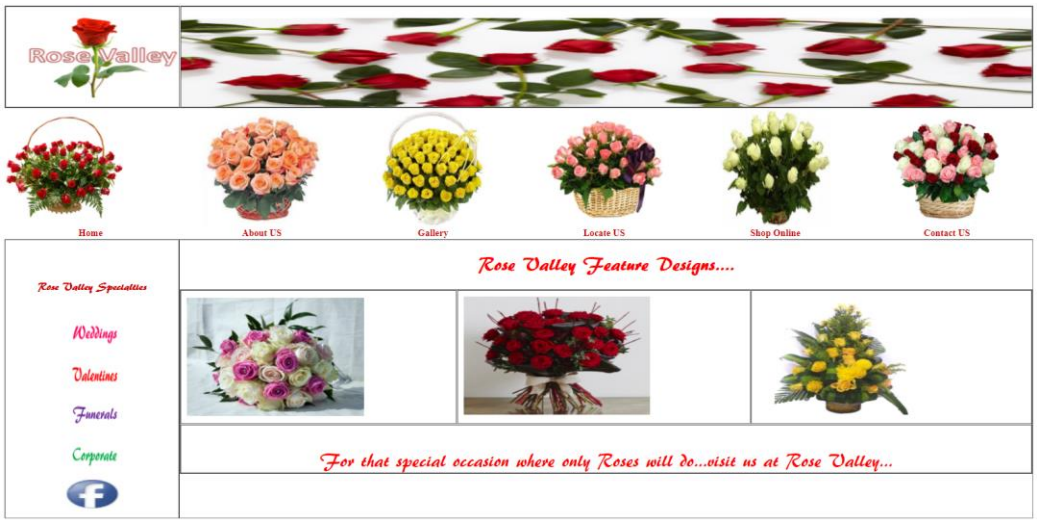
```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title>Rose Valley</title>
<style type="text/css">
.style3
{
color: red;
text-align: center;
}
.style4
{
color: red;
font-family: "AR BERKLEY";
font-size: xx-large;
}
.style8
{
width: 100%;
}
    .style9
    {
        color: red;
    }
    .style10
    {
        font-family: "AR BERKLEY";
        font-size: large;
        color: #CC0000;
    }
</style>
</head>
<body>
<table border="1" style="width: 100%">
<tr>
<td width="17%" height="120" >
<img alt="Rose Valley logo" width="100%" height="150"
src="Images/roseValleyLogo.PNG" />
</td>
<td align="center" colspan = "2">
 <img alt="red rose petals" height="150" src="Images/redRoses.png"
width="100%" /></td>
</tr>
</table>
<table width="100%">
<tr border="1">
```

```
<td width="15%" class="style9"><a href="www.rosevalley.com.fj/home.htm"><img
alt=""
        src="Images/basketRedRoses.jpg" width="200" style="border-width:
0px" /></a></td>
<td width ="15%" class="style3">
    <a href="www.rosevalley.com.fj/about_Us.html">
    <img alt="" src="Images/basket%20of%20orange%20roses.jpg" width="200"
        style="border-width: 0px" /></a></td>
<td width = "15%" class="style3">
    <a href="www.rosevalley.com.fj/gallery.html">
    <img alt="" src="Images/basket%20of%20yellow%20roses.jpg" width="200"
        style="border-width: 0px" /></a></td>
<td width = "15%" class="style3">
    <a href="www.rosevalley.com/locateUs.htm">
    <img alt="" src="Images/basket%20of%20pink%20roses.jpg" width="200"
        style="border-width: 0px" /></a></td>
<td width = "15%" class="style3">
    <a href="www.rosevalley.com/shopOnline.htm">
    <img alt="" height="200" src="Images/basket%20of%20cream%20roses2.jpg"
        style="border-width: 0px; height: 200px" width="200" /></a></td>
<td width = "15%" class="style3">
    <a href="www.rosevalley.com.fj/contactUs.htm">
    <img alt=""
src="Images/basket%20of%20white%20pink%20and%20red%20roses.jpg"
        width="200" style="border-width: 0px" /></a></td>
</tr>
<tr>
<td width="15%" style="text-align: center; font-weight: 700; color: #CC0000"
>Home</td>
    <td width="15%" style="color: #CC0000; font-weight: 700; text-align:
center">About US</td>
    <td width="15%" style="color: #CC0000; text-align: center; font-weight:
700">Gallery</td>
    <td width="15%" style="color: #CC0000; text-align: center; font-weight:
700">Locate US</td>
    <td width="15%" style="text-align: center; color: #CC0000; font-weight:
700">Shop Online</td>
    <td width="15%" style="text-align: center; color: #CC0000; font-weight:
700">Contact US</td>
</tr>
</table>
<table border="1" width="100%" cellspacing="1" cellpadding="0">
<tr>
<td width="17%" rowspan="2">
<h2 align="center">
     </h2>
    <h2 align="center">
        <span class="style10">Rose Valley Specialties</span></h2>
    <p align="center">
        <br/>
        <a href="http://www.rosevalley.com.fj/weddings.htm">
<img alt="pink wedding label" height="50" src="Images/wedding.jpg"
width="90" style="border-width: 0px" /></a><br/><br/>
        <a href="http://www.rosevalley.com.fj/valentines.html">
<img alt="red valentines label" height="50" src="Images/valentines.jpg"
width="90"
```

```
style="align:center" style="border-width: 0px"/></a><br/><br/>
        <a href="http://www.rosevalley.com.fj/funerals.htm">
<img alt="purple funeral label" height="50" src="Images/funerals.jpg"
width="90"
style="text-align: center" /></a><br/><br/>
        <a href="http://www.rosevalley.com.fj/corporate.htm">
<img alt="green corporate label" height="50" src="Images/corporate.jpe"
width="90" /></a><br /><br />
        <a href="http://www.facebook.com/rosevalley_Fiji">
<img alt="facebook icon" height="50" src="Images/like_us_on_facebook.jpg"
width="90"
style="border-width: 0px" /></a></p>
</td>
<td valign="top" colspan="2" >
<h1 align="center" class="style4" >Rose Valley Feature Designs....</h1>
<table class="style8" align="center" cellspacing="2" border="1">
<tr>
<td valign="middle" style="padding: 10px">
<img alt="white and pink roses wedding bouquet" height="200"
        src="Images/wedding_bouquet.jpg" width="300" /></td>
<td valign="middle">
<img alt="red roses valentines bouquet" src="Images/valentines_bouquet.jpg"
        height="200" style="padding: 10px; width: 315px" /> </td>
<td valign="middle">
<img alt="yellow roses modern bouquet " src="Images/corporate1.jpg"
height="200"
        width="300" style="position: relative; padding: 10px; margin: 0px"
/> </td>
</tr>
<tr>
<td table colspan = 3 style="text-align: center">
<span class="style4">
<br />
For that special occasion where only Roses will do...visit us at Rose
Valley...</span></td>
</tr>
</table>
</td>
</table>
</body>
</html>
```

Answer:


Rose Valley
Home
About US
Gallery
Locate US
Shop Online
Contact US
Rose Valley Specialties
Weddings
Valentines
Funerals
Corporate
Rose Valley Feature Designs....
For that special occasion where only Roses will do...visit us at Rose Valley...

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2) Using good coding practices ensures the greatest standards-compliance, presentation, and usefulness of the content. Given the source code below, identify five specific errors, state the XHTML syntax rule that the error violates, and state how to correct it using examples.

[10 Marks]

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
 "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en" lang="en-US">
    <head>
        <TITLE>Transitional DTD XHTML</TITLE>
    </head>
    <body bgcolor= "#dae1ed">
        <div Style="color:#090;font-size:40px;
                font-weight:bold;text-align:center;
                margin-bottom:-25px;">Information Systems<\div>
        <p Style="text-align:center;font-size:20px;">
                A computer science portal
        <p Style="text-align:center;font-size:20px;">
        Option to choose month:
        <select name=month>
            <option selected="selected">January</option>
            <option>February</option>
            <option>March</option>
            <option>April</option>
            <option>May</option>
            <option>June</option>
            <option>July</option>
            <option>August</option>
            <option>September</option>
            <option>October</option>
            <option>November</option>
            <option>December</option>
            </p>
        </select>
        </body>
</html>
```

Error 1 <TITLE>Transitional DTD XHTML</TITLE>

Syntax rule violated All tag names must be in lower case

Correction <title>Transitional DTD XHTML</title>

Error 2 <div Style="color:#090;font-size:40px;
font-weight:bold;text-align:center;
margin-bottom:-25px;">Information Systems<\div>

Syntax rule violated All tags must have a corresponding matching closing tag (container tags)

Correction <div……>

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Error 3 Style -attribute name is starting with a capital letter

Syntax rule violated All tag names must be in lower case

Correction style

Error 4 name=month

Syntax rule violated All attribute values must be enclosed in quotes (either double or single – the notes use double)

Correction name="month" or name= 'month'

Error 5 <select> </select>

Syntax rule violated Nested tags must be terminated in the reverse order to the way they are written.

Correction <select> </select>

Error 6

Syntax rule violated cannot use an inline level style for a block level element with a number of style declorations using the style attribute

Correction

Answer

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
 "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en" lang="en-US">
    <head>
        <TITLE>Transitional DTD XHTML</TITLE>
    </head>
    <body bgcolor= "#dae1ed">
        <div Style="color:#090;font-size:40px;
                font-weight:bold;text-align:center;
                margin-bottom:-25px;">Information Systems <\div>
        <p Style="text-align:center;font-size:20px;">
                A computer science portal
        <p Style="text-align:center;font-size:20px;">
        Option to choose month:
        <select name=month>
            <option selected="selected">January</option>
            <option>February</option>
            <option>March</option>
            <option>April</option>
            <option>May</option>
            <option>June</option>
            <option>July</option>
            <option>August</option>
```

```
            <option>September</option>
            <option>October</option>
            <option>November</option>
            <option>December</option>
            </p>
        </select>
        </body>
</html>
```

3)

(a) Describe 5 options that you can provide on a Web site for navigation options.

[5 Marks]

A simple **text-based navigation bar** can be used to link to the other main pages of the Web site, allowing users to go directly to the pages they want.

- Although the navigation bar lets users access the main pages in the Web site, **links to individual files (external or internal)** can be used to allow users access the exact content they want. This linking method lets the users scroll through the page and then jump to the page they want. The link colors—by default, blue for new and purple for visited—allow users to keep track of which chapters they already have visited.
- In addition to linking to external documents, you also can add **links for navigating within the page itself**. This requires two <a> anchor elements: one uses the name attribute to name a fragment identifier in the document; the other targets the fragment name in the href attribute.
- An internal navigation bar can be used to **link to specific sections within** a single page.
- You can let users **jump** from the current page **to the exact topic they want within different pages**. This requires adding code to both the current page and each page that will be linked to.
- **Page turners** let you move either to the previous or next page in the collection. These work well in a linear structure of pages.

(b) List and describe three issues of concern regarding operating systems and Web design.

[3 Marks]

1 mark for each issue + explanation

Monitors and display software: For many technical and physical reasons, the colors you choose and images you prepare for your site can look vastly different on different machines. Screen resolutions and sizes, color depth, and video hardware and software all affect the look of your Web pages.

Browser versions: Not all browsers are the same on all operating systems. Often software companies release different versions of their browsers based on the popularity of the operating system. For this reason, Internet Explorer is often a release behind on the Macintosh platform. Microsoft does not even make a UNIX version of their browser, although Netscape does. The only solution to this problem is to test your work in as many browsers as possible.

Font choices: Installed fonts vary widely from one computer to another.
Choose fonts that are commonly used; otherwise, the font you choose, if not installed on the user's machine, will appear in a default typeface.

(c) Discuss how navigation bars and Table of Contents differ with respect to links.

[2 Marks]

While the navigation bar lets users access the main pages in the Web site, the table of contents lets users access the exact content they want. The Table of Contents page therefore needs links to the individual chapter files in the Web site.

4)

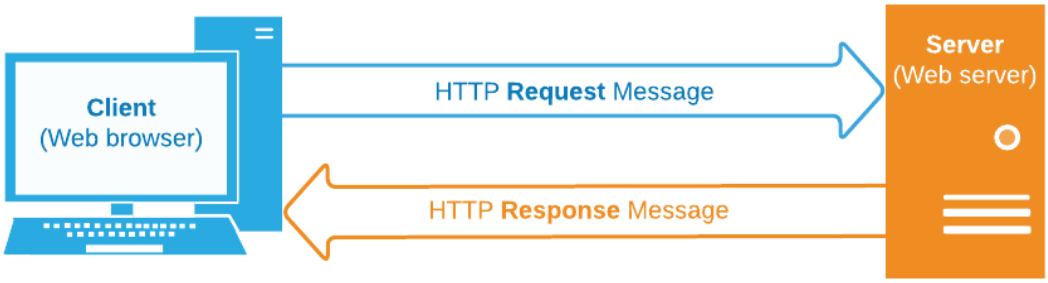
(a) Discuss how you are able to view your Facebook page on your computer. Your discussion should include the core elements of web applications architecture.

[5 Marks]

Client using HTTP/HTTPS protocol requests the Facebook login page. This request is put in an IP envelope and send over the Network. At Facebook's end, the IP envelope is opened and the request was send to one of FB's webservers. The webserver then interacts with the database server to retrieve the requested Facebook member's login page. The database server response with the requested login page to the webser, the webserver then packages this into an IP envelope and sends the response back onto the network to be delivdered to the requesting client.

(b) using a digram illustrate your disction for part (a) above. **[5 Marks]**

HTTP request diagram? full marks can be given when shown data base interactivety as well since they need to be logged in to their facebook account . else 3 marks can be awarded for an image as below



5) Facebook is a website that is very compelling to use, attractive, easy to navigate and a very popular social media website. In designing the Facebook website, the designers took into consideration the following pointers:
   - Crafting the look and feel
   - Design portability
   - Easy access to your information
   - Planning for Clear presentation of information
   - Design for low bandwidth

For each consideration discuss how Facebook's website implemented them.

[10 Marks]

Crafting the look and feel (2 marks)

Grid structure, uniform use of blue and white color scheme and clearly labelled sections etc. afforded a very professional look that is intuitively easy to use.

Design portability (2 marks)

Coding to standards and the use of different stylesheets etc. enable Facebook's site to be accessible on any platform. They also have a mobile version of the main site.

Easy access to your information (2 marks)

Facebook used a great information design that is based on what the users are expected to be able to do and as such structured the main links accordingly hence maintaining the 3-click rule of accessing information.

Planning for Clear presentation of information (2 marks)

Use of the grid structure once again and active use of white space that partitions the various sections of FB with clear headings and icons that are representative of the various activities e.g. mail icon for messaging, etc.

Design for low bandwidth (2 marks)

Minimalistic use of graphics and strategic reusing of universally understood icons.